

Agreement and Surprise: Global Equilibrium from Local Repair

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They do not comprehend how a thing agrees at variance with itself; it is an attunement turning back on itself, like that of the bow and the lyre.

Heracitus, fragment B51, in the translation of Kahn [58]

Abstract

This paper is about one mechanism and the claim that it is at work in every layer of reality. Many parts, each holding a state and able to read only its neighbours, each correcting itself until it agrees with what it reads: the configuration in which all of them agree is the state of the whole, and no part computed it. While the evidence stands, nothing changes at that state. New evidence makes some part disagree; repair spreads from there and, where it converges, reaches a new agreement that contains the evidence. Perfect agreement would also be perfect stillness, so what exists is agreement held off balance by evidence from outside it, a condition we call detuning, and kept in motion by evidence that keeps changing. We define the mechanism and prove what follows from the definition: when an agreement exists, when local correction reaches it and when it circles forever, how averages balance where states do not rest, how far an agreement moves when a part disagrees, and how learning rules of a known class apply the same correction to slower variables. We then find the structure, with the hypotheses each field requires, in the consensus physics of finite observers, in graded and spiking neural networks, in natural selection, in markets and in scientific inquiry. A simulation on the wiring diagram of a fly brain shows the parts of the claim coming apart: states that keep moving, averages that settle, and numerical evidence for more than one equilibrium. The universality is a thesis, supported by one definition and by theorems proved field by field, and we say what would refute it.

1 The thesis

Picture a great many small parts. Each holds a state of its own, each can read the states of a few neighbours and nothing else, and each keeps correcting its own state until it agrees with what it reads. No part needs to represent the whole. When every part agrees with its neighbours, the configuration of all of them is at rest, and that configuration is what the whole system holds: an agreement reached without a central solver.

This paper is about the claim that this picture is the same picture at every layer of reality, and about what can be proved once the picture is written down. In the physics of finite observers the parts are observers who must agree about the records they share. In a brain they are neurons, each settling against the cells it is wired to. In a population they are organisms, in a market traders, in a science the people who hold its beliefs. Three claims attach to the picture.

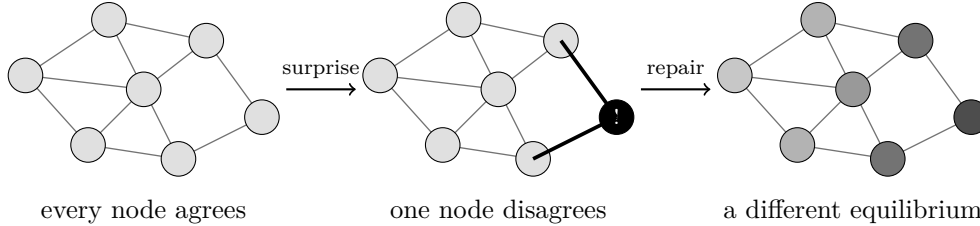


Figure 1: The principle. Nodes read only their neighbours. While every node agrees with what it reads, nothing changes. Evidence at one node makes it disagree, repair spreads along the connections, and the configuration that results contains the evidence. Reaching the right-hand panel needs a condition on the corrections, which Section 3 states; without it the system can circle forever, and some constraints can be impossible to satisfy together.

State. When a system of connected parts is at rest, what it holds is a global agreement made of local conditions. When it moves, it is approaching an agreement, circling one or wandering, and its averages obey the agreement’s balance law.

Change. The agreement changes only when evidence reaches some part that the agreement does not contain, so that the part disagrees. Repair spreads from there and, where it converges, reaches a different agreement that includes the evidence; under a contraction, how far the agreement moves is bounded above and below by how much the part disagreed.

Detuning. Perfect agreement is complete stillness. Standing evidence holds a system off the rest it would reach alone, and the disagreement that evidence creates at the old rest is the system’s detuning. Evidence that keeps changing keeps the system moving between rests; so can dynamics that do not settle.

A part can disagree because it holds evidence the agreement does not account for. In a brain it is a sensory neuron clamped by the world, or an outcome that contradicts a prediction. In a market it is a trader with news. In a population it is a mutant or a changed environment. In a science it is an anomaly. In observer physics it is a record that the other observers have not reconciled. In each case the same question follows: can the local corrections absorb the evidence into a new agreement? Section 3 says when they can, when they circle instead, and when the constraints cannot all be met together.

What is old and what is added. Much of this is old, field by field. Economists proved that market equilibria exist, that price adjustment reaches them under conditions, and that it can cycle around them without those conditions [6, 7, 8]. Evolutionary game theory proved the same three results for selection [5, 14, 16]. Epistemology has theorems on when observers who exchange what they know must come to agree [18, 19]. Numerical analysis solves equations by letting the unknowns correct one another in any order [26], and fixed-point theory, potential dynamics and sheaf theory supply general languages for local-to-global organisation [27, 31, 25]. Two recent programs have the same form. Observer Patch Holography reconstructs physical structure from finite observers who must agree on what they share [43, 44], and Cadence builds learning machines from patches that settle against their neighbours and learn from the difference between a settled state and a disturbed one [56, 57].

This paper puts those results in one frame and asks which hypotheses carry over. The individual theorems are classical; what is added is the common definition, a quantitative statement of how

far an agreement moves when a part disagrees, the account of where each correspondence stops, and a measurement that separates the meanings of agreement. Universality is a thesis motivated by these connections; giving them one name does not make it a consequence.

How to read this paper. Section 2 gives the definition and introduces detuning. Section 3 proves what follows from the definition under stated hypotheses: when an agreement exists, when local correction reaches it, when it can circle forever, how averages agree where states do not, and how far the whole moves when a part disagrees. Section 4 shows how learning rules of a known class apply the same correction to slower variables. Section 5 walks through the layers, from the physics of observers through brains, life, markets and inquiry, and in each case names the hypotheses under which the theorems of Section 3 apply and what the field itself has proved. Section 6 reports a simulation on the wiring diagram of a fly brain in which existence, reachability and uniqueness of the agreement come apart. Section 7 says what would refute the thesis, and Section 8 what it does not claim. Every formal statement is followed by a sentence that says what it means in plain words; a reader who skips the proofs loses nothing of the argument.

Two words. The residual a part holds against its neighbours is called its *disagreement* throughout, and an arrival of evidence that creates disagreement where there was none is a *surprise*. Both are deterministic notions about states, with no reference to subjective experience or to an information measure. The irregular motion that the word chaos denotes in dynamics is, in this vocabulary, motion of the states while the averages balance, and Section 6 shows such motion in a network.

2 Repair networks

The definition has to be small enough to fit a neuron, a price and an observer of a physical screen. It uses four things: the parts, the state each part holds, the neighbours a part can read, and the rule by which a part corrects itself.

Definition 1 (Repair network). A *repair network* consists of a finite set V of nodes and, for each node $i \in V$,

- a state space $X_i \subset \mathbb{R}^{d_i}$, the values node i can hold;
- a neighbourhood $N(i) \subseteq V$, the nodes whose states node i can read;
- a set E_i of evidence that can be imposed on node i from outside;
- a repair map $R_i: \prod_{j \in N(i)} X_j \times E_i \rightarrow X_i$.

The value $R_i(x_{N(i)}, e_i)$ is the state node i must hold in order to agree with what its neighbours hold and with its own evidence. We write $X = \prod_i X_i$, $E = \prod_i E_i$ and $R(x, e) = (R_i(x_{N(i)}, e_i))_{i \in V}$.

In words: a node holds a value, reads some others, and has a rule saying which value would fit what it reads. For a model neuron the state is its potential, its neighbours supply synaptic inputs, and sensory drive is evidence. For a price the repair target is the next price proposed by an adjustment rule; excess demand can depend on all prices, so this node reads widely. An observer patch repairs its record against shared interfaces, but a record satisfying every interface need not exist. The map R_i takes no argument outside $N(i)$ and E_i . The definition allows $N(i) = V$, so locality in a substantive sense is a further commitment: neighbourhoods or communication costs have to stay limited as the system grows.

Definition 2 (Disagreement and equilibrium). Fix norms on the spaces $\mathbb{R}^{d_i}$ and a norm $\|\cdot\|$ on their product. The *disagreement* of node i in state x under evidence e is $\delta_i(x, e) = \|R_i(x_{N(i)}, e_i) - x_i\|$, and the disagreement of the network is $D(x, e) = \|R(x, e) - x\|$. A state x^* is an *equilibrium* under e when no node disagrees:

$$x^* = R(x^*, e).$$

In words: a node disagrees when the value it holds differs from the value that would fit its neighbours, and an equilibrium is a configuration in which every node holds the fitting value. The equilibrium is defined by $|V|$ local conditions. It is global in the sense that all of them hold at once; no separate global solver is required.

Definition 3 (Repair). *Repair* is any process in which nodes move toward their repair targets using what they can read. Four forms recur:

$$\begin{aligned} \text{synchronous: } & x^{t+1} = R(x^t, e), \\ \text{damped: } & x^{t+1} = x^t + \eta (R(x^t, e) - x^t), \quad 0 < \eta \leq 1, \\ \text{relaxation: } & \dot{x} = R(x, e) - x, \\ \text{asynchronous: } & \text{at each step some nodes apply their map, possibly to stale readings.} \end{aligned}$$

In words: the nodes may all jump to their targets at once, move part of the way, flow continuously, or act whenever they like on whatever they last read. The first three processes have exactly the equilibria as their rest points; damped repair stays in a convex domain, and continuous repair is treated as a deterministic flow, which local Lipschitz continuity of R in the state secures (the existence theorem below needs only continuity). For asynchronous repair every node must act eventually and stale readings must be refreshed eventually, since a schedule that ignores a node can hide its disagreement, and a message in flight can disturb a node that agrees; the pending messages are then part of the state that has to settle. The processes differ in whether they reach a rest point, which is the subject of Section 3.

Surprise and slow variables. A *surprise* is a change of evidence $e \rightarrow e'$ under which the current equilibrium stops being one, $D(x^*(e), e') > 0$. A *slow variable* is a parameter θ of the repair maps, $R = R_\theta$, that changes on a longer time scale than the states: a synaptic weight, a gene, an institution, a theory. The states settle for fixed θ ; learning and evolution move θ . The separation of time scales is a modelling assumption, and Section 4 says what it buys.

Detuning. Evidence that stays, such as a constant sensory input, moves a network's resting point. Choose a reference condition, call it 0, with an equilibrium $x^*(0)$ (one of them, if there are several). The *detuning* by evidence e is the disagreement that e creates at that reference rest,

$$\Delta(e) = D(x^*(0), e) = \|R(x^*(0), e) - x^*(0)\|.$$

In words: detuning asks how much the evidence disagrees with the old rest. If the map under e contracts, Theorem 12 places the new rest between $\Delta/(1+q)$ and $\Delta/(1-q)$ from the reference one, so the offset of the equilibrium is proportional to the detuning, and when $\Delta = 0$ the two rests coincide.

A shifted rest is a rest. The simplest example is $R(x, e) = e$ on $[-1, 1]$: its detuning is $|e|$, and at $x = e$ nothing moves. A contracting network settles under any constant input; what keeps a system in motion is evidence that keeps changing, or dynamics that do not settle, as the two-node loop of Section 3 shows. The systems of Table 1 live on streams of evidence, and Section 5 names the stream in each layer.

Table 1 lists the systems treated below in these terms.

System	Node	Repair	Equilibrium	Surprise	Slow variable
Observer physics	observer patch	reduce mismatch on shared interfaces	public normal form; consistent if repair is complete	a record the others have not reconciled	carrier architecture, held fixed
Graded neural network	neuron or patch	state follows its inputs or a local energy derivative	settled activity, the world model in use	clamped input, outcome, or a higher region's nudge	synaptic weights
Spiking network	neuron	integrate spikes, fire, reset	stationary firing rates, where attained	stimulus change	synaptic weights
Evolution	type or individual	frequency follows payoff against those it meets	rest point of selection	mutant, changed environment	genome, the game itself
Market	trader or market	price follows excess demand	prices at which markets clear	news	technology, institutions
Science	researcher	credence follows the evidence colleagues report	consensus	anomaly	theory

Table 1: One vocabulary, six systems. Section 5 gives the definitions, the hypotheses and the theorems behind each row; the last column names what each field treats as its slow variable, and no row asserts that the field has a proved law for how it moves.

3 What follows from the definition

Three questions can be asked of any repair network. Does an agreement exist? Does repair reach it? Is it the only one? The answers come from different hypotheses, and Section 6 shows a measured network in which they come apart. Each statement below is followed by what it says in words.

3.1 An equilibrium exists

Theorem 4 (Existence). *Let every X_i be nonempty, compact and convex, and let $R(\cdot, e)$ be continuous. Then the network has an equilibrium under e .*

Proof. The product X is nonempty, compact and convex, and $R(\cdot, e)$ maps it continuously into itself. Brouwer's fixed-point theorem [1] gives $x^* = R(x^*, e)$. $\square$

In words: if each node's allowed values form a nonempty closed, bounded, convex set, and correction varies continuously, there is a configuration in which every node agrees with its repair target. These conditions cover many models, but they cannot simply be dropped: $R(x) = x/2$ has no fixed point in the open interval $(0, 1)$, and a nontrivial rotation of a circle has none. When a node may hold any of a convex set of best answers, Kakutani's theorem [2] replaces Brouwer's when the correspondence is upper hemicontinuous with nonempty compact convex values on the same compact convex domain. Nash's equilibrium of a game [4] and the Arrow–Debreu equilibrium of an economy [6] are instances (Section 5). The theorem is silent on how to find the equilibrium and on how many there are.

3.2 Repair reaches it

Theorem 5 (Reachability under contraction). *Let X be nonempty, closed and convex and let $R(\cdot, e)$ be a contraction: $\|R(x, e) - R(y, e)\| \leq q \|x - y\|$ with $0 \leq q < 1$. Then*

1. *the equilibrium $x^*(e)$ exists and is unique;*
2. *synchronous and damped repair converge to it from every start, with $\|x^t - x^*\| \leq (1 - \eta + \eta q)^t \|x^0 - x^*\|$;*
3. *relaxation converges to it, with $\|x(t) - x^*\| \leq e^{-(1-q)t} \|x(0) - x^*\|$;*
4. *if the evidence lies in a normed space, $x^*(e')$ is an equilibrium under other evidence e' , and $\|R(x, e) - R(x, e')\| \leq L \|e - e'\|$ for all x , then $\|x^*(e) - x^*(e')\| \leq L \|e - e'\| / (1 - q)$.*

Proof. Item 1 is Banach's theorem [3]. For item 2, the damped map $x \mapsto (1 - \eta)x + \eta R(x, e)$ has the same fixed point and Lipschitz constant $1 - \eta + \eta q < 1$. For item 3 put $z = x - x^*$; then $\dot{z} = -z + R(x, e) - R(x^*, e)$ and the upper right derivative of $\|z\|$ is at most $-\|z\| + q\|z\|$. For item 4, $\|x^*(e) - x^*(e')\| \leq \|R(x^*(e), e) - R(x^*(e'), e)\| + \|R(x^*(e'), e) - R(x^*(e'), e')\| \leq q \|x^*(e) - x^*(e')\| + L \|e - e'\|$. $\square$

In words: if one round of corrections always brings two configurations closer together, then there is exactly one agreement, and synchronous, damped and continuous repair approach it. Under the stated evidence bound, a small change in evidence moves it by a small amount. The number q bounds how much a round of corrections leaves of a difference; the closer the bound is to one, the weaker the guarantees on speed and sensitivity. When the contraction holds in a weighted maximum norm, asynchronous repair with bounded delays converges as well, provided every node acts infinitely often and old readings are eventually discarded. Chazan and Miranker proved this for linear maps under the name chaotic relaxation [26], and Bertsekas extended it to nonlinear ones [27]. No clock and no fixed order of the nodes is needed; fairness is.

A contraction is a strong hypothesis. A second route asks instead that the nodes all pull on one quantity.

Theorem 6 (Equilibrium limit points under a potential). *Let every X_i be nonempty and compact and let $\Phi: X \rightarrow \mathbb{R}$ be continuous. Suppose that for every node i and every state of the other nodes the function $x_i \mapsto \Phi(x_i, x_{-i})$ has a unique minimiser on X_i , and that this minimiser is the repair target $R_i(x)$. Let the nodes repair one at a time in a fixed cyclic order. Then Φ never increases along the sequence of states, and every limit point of the sequence is an equilibrium.*

Proof. Each single repair replaces x_i by the minimiser of $\Phi(\cdot, x_{-i})$, so Φ does not increase. Being bounded below on the compact set X , the values $\Phi(x^k)$ at the ends of successive cycles converge to some Φ_∞ . Uniqueness of the minimiser and compactness make each R_i continuous. Let $x^k \rightarrow \bar{x}$ along a subsequence and let $y^k = (R_1(x^k), x_{-1}^k)$ be the state after the first repair of cycle k . Then $\Phi(x^{k+1}) \leq \Phi(y^k) \leq \Phi(x^k)$, both ends tend to Φ_∞ , and $y^k \rightarrow (R_1(\bar{x}), \bar{x}_{-1})$. Hence $\Phi(R_1(\bar{x}), \bar{x}_{-1}) = \Phi_\infty = \Phi(\bar{x})$, and by uniqueness of the minimiser $R_1(\bar{x}) = \bar{x}_1$. The state after the first repair therefore also tends to $\bar{x}$ along the subsequence, and the same argument applies to the second node and to each node in turn. For any convergent subsequence sampled inside cycles, pass to one fixed phase and then, by compactness, to convergent cycle starts. The argument just given makes all phases converge to that same equilibrium, so the conclusion holds between cycle boundaries too. $\square$

In words: if each cyclic correction uniquely minimises one shared quantity with respect to its own node, repair runs downhill and whatever it approaches is a configuration that no single node can improve. Such a state need not be the lowest possible, and the theorem does not say that the sequence converges to one point. When every term of Φ containing x_i has its other variables in $N(i)$, the minimiser reads only that neighbourhood and repair is local. Symmetric energy models [28, 29, 30] and potential games [31] are the standard examples; the conclusion needs the unique coordinate minimisers and the cyclic order as stated. Selection has a potential of a different form, derived in Section 5.3.

3.3 Repair can fail to reach it

Proposition 7 (Existence without reachability). *Let two nodes each hold a number in $[-1, 1]$, with repair maps*

$$R_1(x_1, x_2) = \text{clip}(2x_1 - 2x_2), \quad R_2(x_1, x_2) = \text{clip}(2x_1 + 2x_2),$$

where clip truncates to $[-1, 1]$. The network satisfies the hypotheses of Theorem 4. Its only equilibrium is $(0, 0)$. Every relaxation trajectory that does not start at $(0, 0)$ converges to a periodic orbit, and synchronous repair from any other start does not converge either.

Proof. If neither target is truncated, $x_1 = 2x_1 - 2x_2$ and $x_2 = 2x_1 + 2x_2$ force $x_1 = x_2 = 0$. If $x_1 = 1$ with $2 - 2x_2 \geq 1$, then $x_2 = \text{clip}(2 + 2x_2)$ has no solution with $x_2 \leq \frac{1}{2}$; the case $x_1 = -1$ is its mirror image. If $x_2 = \pm 1$ with x_1 not truncated, then $x_1 = 2x_1 \mp 2$ gives $|x_1| = 2$. So $(0, 0)$ is the only equilibrium. The relaxation $\dot{x} = R(x) - x$ has Jacobian $\begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$ there, with eigenvalues $1 \pm 2i$: a source. Trajectories stay in the square, since $|R_i| \leq 1$. In a small disk the vector field is linear and the radius grows strictly, so no nonconstant trajectory has the source in its omega-limit set, and the planar Poincaré–Bendixson theorem makes that compact limit set a periodic orbit. For synchronous repair the linearisation at the origin has eigenvalues $2 \pm 2i$ of modulus larger than one, and only the origin is mapped to the origin, so no other sequence of states converges to it. $\square$

In words: an agreement can exist and be unreachable. The two nodes excite themselves, the first excites the second, and the second inhibits the first: a pursuer and a flee, the smallest excitatory–inhibitory loop, related to the oscillatory models of Wilson and Cowan [35]. Computed numerically, the orbit is the same from every start tested, has period 7.58 in units of the relaxation time, and stays at distance at least 0.84 from the equilibrium. Existence is a theorem about the equations; reaching the equilibrium is a property of the dynamics, and the two can part company in a network of two nodes.

3.4 Averages agree where states do not

Proposition 8 (Agreement on average). *Let $x(t)$ be a relaxation trajectory in a bounded set X , and write $\bar{x}_T = \frac{1}{T} \int_0^T x(t) dt$ and $\bar{R}_T = \frac{1}{T} \int_0^T R(x(t), e) dt$. Then*

$$\|\bar{x}_T - \bar{R}_T\| \leq \frac{\text{diam } X}{T}.$$

Proof. Integrating $\dot{x} = R(x, e) - x$ over $[0, T]$ gives $x(T) - x(0) = T(\bar{R}_T - \bar{x}_T)$. $\square$

In words: whatever the states do, each node’s average value equals the average of what its neighbours asked of it, up to a term that vanishes as the window grows. Even a network that never rests agrees on average. The identity constrains averages; it does not say that the averages

converge, and it does not allow a nonlinear repair map to be pulled through the average. For the damped discrete process the corresponding exact identity is

$$\frac{1}{N} \sum_{k=0}^{N-1} (R(x^k, e) - x^k) = \frac{x^N - x^0}{\eta N},$$

with the same conclusion and the boundary bound $\text{diam } X/(\eta N)$. Two consequences matter below.

Corollary 9 (Linear repair). *If X is also closed and convex and $R(\cdot, e)$ is affine, every limit point of the time averages $\bar{x}_T$ is an equilibrium.*

Proof. For affine R , $\bar{R}_T = R(\bar{x}_T, e)$, so $\|\bar{x}_T - R(\bar{x}_T, e)\| \rightarrow 0$. $\square$

Corollary 10 (Rate law). *Let each node integrate a weighted sum of signals emitted by its neighbours, $R_i(x, e) = \sum_j W_{ij} s_j(x_j) + e_i$, with bounded signals s_j . Then along any relaxation trajectory the average potentials and the average signals satisfy*

$$\bar{x}_i = \sum_j W_{ij} \bar{s}_j + e_i + O(1/T).$$

In words: when nodes communicate by signals, the average potentials and the average signals obey the wiring's linear law up to the finite-window term, and exactly at any joint limit of those averages. What the averages do not supply is the link between the two, $\bar{s}_j = s_j(\bar{x}_j)$: the average of a signal is in general different from the signal of the average. For a Lipschitz repair map on a convex domain the gap is explicit,

$$\|R(\bar{x}_T, e) - \bar{x}_T\| \leq \frac{\text{diam } X}{T} + \frac{L}{T} \int_0^T \|x(t) - \bar{x}_T\| dt,$$

by adding and subtracting $\bar{R}_T$, and persistent oscillation need not make the second term small. A network that never rests can balance its average inputs and outputs without those averages being a fixed point of its original repair map. In the symmetric example of Proposition 7 the computed time average of the orbit is the equilibrium itself to five decimal places; that is a feature of the example's symmetry and no consequence of boundedness alone.

Nodes that emit spikes obey an analogous balance, with resets included.

Proposition 11 (Charge balance). *Let node i carry a potential u_i and an input current g_i , with positive time constants τ_m, τ and locally finite spike trains, obeying*

$$\tau_m \dot{u}_i = -u_i + g_i, \quad \tau \dot{g}_i = -g_i + \tau \sum_j w_{ij} \sum_k \delta(t - t_j^k),$$

where t_j^k are the spike times of the nodes j that node i reads, external sources of spikes among them. Let node i emit a spike whenever u_i reaches a positive threshold ϑ , and let a spike reset u_i to zero. If potentials and currents stay bounded, the firing rates ν_i and the finite-window average potentials $\bar{u}_i$ satisfy, up to $O(1/T)$,

$$\tau_m \vartheta \nu_i = \tau \sum_j w_{ij} \nu_j - \bar{u}_i.$$

Proof. Integrate the two equations over $[0, T]$ and divide by T . The current gains w_{ij} at each spike of node j , and each spike of node i removes ϑ from its potential. With the endpoint terms written out,

$$\bar{g}_i = \tau \sum_j w_{ij} \nu_j - \frac{\tau}{T} (g_i(T) - g_i(0)), \quad \tau_m \vartheta \nu_i = \bar{g}_i - \bar{u}_i - \frac{\tau_m}{T} (u_i(T) - u_i(0)),$$

and boundedness makes both endpoint terms $O(1/T)$. $\square$

In words: a neuron that fires is a node whose average obeys a linear law set by the wiring, with one local unknown, its mean potential. A spike records that the sender's potential reached its threshold, and the law holds whatever the timing of individual spikes. The law constrains rates; it does not by itself determine them from the wiring or prove that they become stationary. A silent node has $\nu_i = 0$ and its average potential approaches its average input. Nodes that hold their state during a refractory period or discard pending current at a spike add local terms of their own to the balance.

3.5 The equilibrium moves as much as the nodes disagree

Theorem 12 (Equilibrium displacement is bounded by disagreement on both sides). *Let $R(\cdot, e')$ have a fixed point $x^*(e')$ and be a contraction with constant $0 \leq q < 1$. Let x be any state, and let $D = \|R(x, e') - x\|$ be the disagreement at x under evidence e' . Then*

$$\frac{D}{1+q} \leq \|x^*(e') - x\| \leq \frac{D}{1-q}.$$

Proof. Write $x^* = x^*(e')$. Then $\|x^* - x\| \leq \|R(x^*, e') - R(x, e')\| + \|R(x, e') - x\| \leq q\|x^* - x\| + D$, and $D \leq \|R(x, e') - R(x^*, e')\| + \|x^* - x\| \leq (1+q)\|x^* - x\|$. $\square$

In words: take x to be the agreement that held before new evidence arrived. If no node disagrees with the evidence, $D = 0$ and nothing moves. If some node disagrees, the agreement moves, by an amount between $D/(1+q)$ and $D/(1-q)$. Change in the whole is caused by disagreement and is bounded in proportion to it. The two inequalities are the standard residual estimates for a contraction; their content here is a quantitative link between the residual created by new evidence and the motion of the whole. The factor $1/(1-q)$ bounds amplification: when q is well below one, it limits the overall response to a small surprise; near $q = 1$, it permits a larger response without guaranteeing it. It says nothing about how far through the graph a disturbance travels, which needs bounds on the couplings.

3.6 There can be more than one

Neither Theorem 4 nor Theorem 6 gives uniqueness. Without a contraction a network can have several equilibria, and when repair converges, the one it reaches can depend on where it started and on the order of the corrections. History enters here: the state before a surprise selects the agreement after it, and two systems with the same parts, the same rules and the same present evidence can differ, the difference being their past. Section 6 reports two distinct numerical equilibria of one network under one stimulus.

4 Learning is slow repair

When repair converges, it settles for fixed repair maps. Most of the systems in Table 1 also change their maps: synapses change, genomes change, theories change. The claim of this section is that one event drives both: a surprise moves the fast state to a new agreement, and the difference between the agreement before and the agreement after tells the slow variables how to change. Precise versions are available for the energy-based models below; no learning law follows from the definition of a repair network by itself.

For energy-based learning, let the slow variables be θ , let the states settle at stationary points of an energy $E(\theta, x)$, and let $C(x)$ measure the disagreement between a settled state and an outcome

imposed from outside. Define $F(\theta, \beta, x) = E(\theta, x) + \beta C(x)$. Assume E and C are twice continuously differentiable and that a selected stationary branch x_θ^β of $F(\theta, \beta, \cdot)$ is continuously differentiable in (θ, β) near $(\theta_0, 0)$; a nonsingular state Hessian supplies such a branch by the implicit-function theorem. The state x_θ^0 is the *free* equilibrium and x_θ^β the *nudged* one, pushed toward the outcome with strength β . Following that branch by relaxation also requires its local stability.

Theorem 13 (Two equilibria give the gradient; Scellier and Bengio [32]). *Under these assumptions the loss $J(\theta) = C(x_\theta^0)$ satisfies*

$$\frac{dJ}{d\theta} = \lim_{\beta \rightarrow 0} \frac{1}{\beta} \left(\frac{\partial E}{\partial \theta}(\theta, x_\theta^\beta) - \frac{\partial E}{\partial \theta}(\theta, x_\theta^0) \right).$$

Proof. Let $G(\theta, \beta) = F(\theta, \beta, x_\theta^\beta)$. Stationarity in x removes the derivatives through the branch, so $dG/d\beta = C(x_\theta^\beta)$ and $dG/d\theta = \partial E/\partial \theta(\theta, x_\theta^\beta)$; the assumptions make both continuously differentiable. Equality of the mixed second derivatives gives $\frac{d}{d\theta} C(x_\theta^\beta) = \frac{d}{d\beta} \frac{\partial E}{\partial \theta}(\theta, x_\theta^\beta)$, and at $\beta = 0$ the left side is $dJ/d\theta$. $\square$

In words: to know how every slow variable should change so that the settled state agrees better with the world, the network needs two settled states, one free and one gently pushed toward the outcome. For the pairwise energy below, each connection needs only what it can read at its own two ends in those states. For a symmetric pairwise energy $E = -\sum_{i < j} \theta_{ij} s_i s_j + \dots$, with one parameter per undirected pair, $\partial E/\partial \theta_{ij} = -s_i s_j$, and the contrast

$$\Delta \theta_{ij} = \frac{\eta}{\beta} \left((s_i s_j)^{\text{nudged}} - (s_i s_j)^{\text{free}} \right)$$

approaches a negative-gradient step as $\beta \rightarrow 0$. No separate backward pass is needed: the outcome disturbs the equilibrium, its influence spreads through the recurrent couplings, and each connection moves by the local difference it sees. At a point with nonzero gradient, a sufficiently small nudge and then a sufficiently small learning rate give descent; finite nudges, incomplete settlement, a change of branch or a large learning rate have no such guarantee.

Two speeds, one mechanism. When both states and parameters descend an energy, their idealised flows have the form

$$\dot{x} = -M_x \nabla_x F, \quad \dot{\theta} = -\varepsilon M_\theta \nabla_\theta F, \quad 0 < \varepsilon \ll 1,$$

with positive semidefinite mobility matrices, and for fixed boundary data

$$\dot{F} = -\nabla_x F^\top M_x \nabla_x F - \varepsilon \nabla_\theta F^\top M_\theta \nabla_\theta F \leq 0.$$

This is one precise meaning of repair at two speeds: fast and slow variables both lower the declared objective. Lowering the joint energy does not by itself lower the outcome loss at the next settlement; equilibrium propagation supplies that gradient from two fast equilibria. The two descriptions need their own objective and update rule, and neither follows from calling a parameter slow. Selection in Section 5.3 has the same dissipative form with a different objective and a different mobility.

No surprise, no learning. If $\nabla_x C(x_\theta^0) = 0$, the free state is also a stationary point of the nudged energy, and when the local branch is unique the two states coincide and every $\Delta \theta_{ij}$ is zero. A network governed by this rule learns only where it is surprised: agreement with the outcome at a minimum of the discrepancy leaves every slow variable where it was. This is the lesson of Theorem 12 one level up, proved for this rule from the stationary branch; biological plasticity, drift and mutation are covered where this rule describes them.

Directed interactions. The theorem needs an energy and the corresponding state dynamics. Ordinary pairwise gradient models have reciprocal interactions; a directed predictive graph defines an energy when its repair follows one. Measured nervous systems do not automatically supply such dynamics. For general vector-field dynamics the two-state contrast is in general different from the loss gradient [33], and the discrepancy is controlled by the antisymmetric part of the network’s Jacobian at equilibrium under local assumptions [34]. How closely a directed wiring follows the gradient is a matter of measurement, and no measurement in this paper settles it.

5 The structure in the layers of reality

The definition and the theorems of Sections 2 to 4 say nothing about any particular system. This section walks through the layers and does the same four things in each: it says where the nodes are and what repair is, it names the hypothesis under which the theorems apply and what the field has proved on its own, it names what can disturb or shift its agreements, and it marks where the correspondence stops. The order runs from the smallest claim about what exists to the largest: the physics of observers, brains, life, markets, inquiry, and the computations that use the mechanism as a tool.

5.1 Physics: observers who must agree

Observer Patch Holography (OPH) asks how public physical structure can arise when finite observers must agree about the records they share [43, 44]. Its consensus layer is a repair network with finite state spaces, and that layer is what this subsection takes first. The program’s reconstruction then adds a declared carrier and operational structure: twelve boundary ports with icosahedral incidence, typed overlaps, a complete reversible response, and maximal randomness relative to a declared reference subject to agreement. Those additions turn a general account of settlement into particular statements about geometry, gauge symmetry, quantum probability, gravity and dissipation. We keep the two levels apart: what the consensus layer proves about any such network, and what the architecture adds under premises of its own.

Definition 14 (Patch net [44]). Let $G = (V, E)$ be a finite connected graph. Each vertex i is an observer patch with a finite state space S_i , and the global state space is $\Sigma = \prod_i S_i$. Each edge $e = \{i, j\}$ carries an interface alphabet I_e and two projections $\pi_{i,e}: S_i \rightarrow I_e$ and $\pi_{j,e}: S_j \rightarrow I_e$. A state s is consistent on e when $\pi_{i,e}(s_i) = \pi_{j,e}(s_j)$, and the consistency set C is the set of states consistent on every edge. With weights $w_e > 0$ and mismatch scores $d_e \geq 0$ that vanish exactly on equal arguments, the inconsistency potential is

$$\Phi(s) = \sum_{e=\{i,j\}} w_e d_e(\pi_{i,e}(s_i), \pi_{j,e}(s_j)), \quad s \in C \iff \Phi(s) = 0.$$

In words: an observer holds a record, two observers who share an interface must show the same thing on it, and the potential counts the disagreements. A state with $\Phi = 0$ is a world every observer agrees on. Each observer reads only its shared interfaces, and a repair is a local move accepted when it lowers the mismatch on the interfaces it touches.

Public facts from local repair. On the strictly descending branch every accepted move lowers the declared measure, $\Phi(t) < \Phi(s)$, and since the measure takes finitely many values every repair sequence is finite [44]. This is a finite descent argument alongside the coordinate-minimisation result of Theorem 6. If the repair relation on the observable quotient is terminating and locally confluent,

Newman’s lemma gives each initial quotient state a unique reachable normal form, and under repair completeness, the further condition that normal forms are exactly the consistent states, that endpoint is an agreement state. Any observable defined on the quotient then takes the same value whatever the schedule, and the consensus paper draws the consequence in these words: “If physical law is identified with this quotient-observable map, then physical law is objective.” In words: what the observers end up agreeing on does not depend on who repaired first. Confluence gives uniqueness among the normal forms reachable from one initial state; contraction in Theorem 5 gives one fixed point for the whole domain. Different initial states can have different consensus normal forms.

Loops can forbid full agreement. For abelian edge demands $x_v - x_u = b_e$ a global assignment exists exactly when every signed cycle sum vanishes. The triangle over $\mathbb{Z}_2$ with demands 0, 0, 1 has none: every edge can be satisfied alone and the three cannot be satisfied together, the minimum of the mismatch is positive, and in the paper’s words “every minimizer contains irreducible residual inconsistency” [44]. In words: an equilibrium is a state that no local repair improves; agreement on every connection is a stronger property, and the constraints around its loops can exclude it. Finite descent and the cycle obstruction are both machine-checked [44, 53].

Noise. For repair disturbed by noise the consensus paper proves the following bound. If fair blocks of repair uniformly contract the expected distance D to the set of normal forms, $\mathbb{E}[D(\tilde{B}_m(q))] \leq \lambda D(q) + \varepsilon$ conditional on each starting state, with $0 \leq \lambda < 1$, $\varepsilon \geq 0$ and finite initial expected distance, then $\limsup_m \mathbb{E}[D(q_{\tau_m})] \leq \varepsilon/(1 - \lambda)$ [44]. The factor $1/(1 - \lambda)$ has the same geometric-series form as $1/(1 - q)$ in Theorems 5 and 12.

Why anything happens: detuning. A settled public record is fixed under its repair rule, and by Theorem 12 nothing in it moves while no observer disagrees. OPH states the same from the side of physics: “A screen at exact self-similar balance carries no events, so records require detuning” [45], and “exact balance is too symmetric to support durable records, structure, and dynamics” [46]. Its detuning law fixes the size of the departure. The ratio P that characterises a cell of the screen sits off the self-similar balance point, the golden ratio φ , by an amount set by the fine-structure constant α ,

$$P = \varphi + \sqrt{\pi} \alpha,$$

and the displacement $P - \varphi$ “is what becomes the electromagnetic observation strength” [47]. In words: the world’s records exist because the screen is held off perfect balance, and the electromagnetic coupling is the size of that offset. This is the detuning of Section 2 measured in the screen’s own variables, where $\Delta(e)$ measures the same kind of departure as a residual of a repair map; identifying the two is part of the physical proposal. The source presents the detuning law as a declared principle and the link to the laboratory constant as a premise that needs its own bridge, and it is cited here with that standing.

What the architecture adds. The flagship reconstruction supplies the carrier and asks what agreement forces on it. Five results bear on the thesis. Each is stated with its antecedents and without the detail of its proof; the sources carry both, and the machine-checked parts are named there [43, 53].

Geometry. With the twelve-port adjacency A and the uniform repair mean $T = I - (5I - A)/60$, the normalised long-time response to a complete equally weighted census of centered probes converges to a Gram matrix $G = 4P_{\text{slow}}$ of rank three, where P_{slow} projects onto the slowest nonconstant band of repair. After quotienting by the invisible antipodal-even loads, the signed

integer port records complete, in the metric G selects, to a three-dimensional Euclidean space on which carrier symmetries act as isometries and accumulated records as translations [43, 45]. In words: three dimensions are what is left of a repair response once the fast modes have died away. The normalisation and the infinite-step limit are essential, and reading the metric as physical position needs scale, distinguishability and gluing across carriers that the theorem does not supply.

Gauge symmetry. Assume a complete injective real-linear response derivative D from the twelve-dimensional port representation into the anti-Hermitian operators of a finite-dimensional complex Hilbert space, with image closed under commutators. Assume every proper A_5 carrier recharting a has an implementer U_a satisfying $\text{Ad}_{U_a} D(v) = D(a \cdot v)$, in the response-generated connected group up to a factor centralising the image. Then $D(P_{12}) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2)$ [43, 50]. In words: a twelve-dimensional agreement response that implements its own symmetries has, under these premises, the local Lie-algebra type of the Standard Model. The premises carry weight, and the physical current, the coupling and the global group are further steps.

Quantum weights. On all effects $0 \leq E \leq I$ of the complex matrix algebra $M_d(\mathbb{C})$, an assignment $v(E) \in [0, 1]$ with $v(I) = 1$ that is additive whenever E, F and $E + F$ are effects has the form $v(E) = \text{Tr}(\rho E)$ for a unique density operator, in dimension two as well: the finite Busch–Gleason theorem [54], formalised in the OPH library [49]. In words: once weights have to agree across every context in which effects coexist, they take the Born form. The complex algebra and the additivity across contexts are premises that agreement on a finite set of observed tests does not supply.

Gravity. A symmetric tensor X on a four-dimensional Lorentzian space satisfying $X(k, k) = 0$ for every null vector k is a multiple of the metric. If a realised balance $G_{ab}k^a k^b = \kappa T_{ab}k^a k^b$ holds for every null k on a connected C^3 Lorentzian manifold with a conserved C^1 stress tensor and constant $\kappa = 8\pi G_N$, the contracted Bianchi identity makes the remaining scalar constant and $G_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab}$ follows [43, 51], in the spirit of Jacobson’s equation of state [55]. In words: agreement of curvature and stress in every null direction determines the field equation up to a cosmological constant. The finite tensor step is machine-checked; the smooth statement is a paper argument whose antecedent, one refinement tower realising geometry, stress and entropy balance together, is open.

Dissipation. For a finite state space with a strictly positive reference π and a conditional repair kernel K that resamples from π conditioned on the fibre of the protected public datum, the relative entropy of a strictly positive distribution p splits exactly, $D(p||\pi) = D(p||q) + D(q||\pi)$ with $q = pK$, and the mean entropy production defined by the source’s forward/reverse path ratio equals $D(p||q)$ [43, 52]. In words: the relative information erased by this repair is exactly the disagreement it removes. The kernel is idempotent, so sustained dissipation needs further dynamics, and a fixed public record is compatible with continuing microscopic motion, a point that Section 6 meets again in a brain.

Scope. The consensus paper states its own limit: on its own, the consensus layer fixes neither a Lorentzian metric nor a field equation, and geometric and dynamical content enters through the structure above [44, 43]. The flagship lists five obligations that none of the finite results discharges: physical position and scale, clock and energy calibration, a physical gauge current, one common refinement tower, and a source-selected action; and its universe-level equation $T(\mathfrak{U}) = \mathfrak{U}$ is a closure hypothesis without an existence or uniqueness proof. What this layer contributes to the thesis is this: a shared record is a normal form of local repair between observers, independent of the order of repair from each initial state under the stated hypotheses, with microscopic motion allowed beneath it; the record exists because the screen is detuned; and a specified architecture on this consensus layer reconstructs substantial physical structure under premises that are in view.

5.2 Brains: settlement in graded and spiking networks

A nervous system is the layer at which the principle can be built, proved and measured. This subsection treats a network designed on the principle, then spiking neurons; Section 6 puts both on the wiring of an animal. What detunes a brain is its senses: a standing input shifts the equilibrium, and a changing input keeps the network moving from one equilibrium toward the next.

5.2.1 Graded networks: the equilibrium as world model

Cadence is a design for machines that learn from experience, and in its population model inference and learning are both settlement [56, 57]. Its unit is the patch, “a bounded predictor that holds one state, reads a declared set of signals through its ports, predicts its own state from them, and carries the difference as an explicit error” [56]. With incoming signals u_{ij} , which are sensory samples or the states of other patches, patch i has

$$p_i = \tanh\left(b_i + \sum_j w_{ij}u_{ij}\right), \quad e_i = x_i - p_i, \quad E = \frac{1}{2} \sum_i e_i^2 + \frac{\lambda_x}{2} \|x\|^2.$$

Agreement in this network “means consistency with the learned local predictions, not equality of all patch activities”. The design tests the hypothesis that “a continuing brain’s equilibrium is its world model in operation”, and its account of computation is the second claim of Section 1: “An observation or an outcome that contradicts a prediction disturbs the equilibrium, and repairing it is the computation” [56]. In words: the settled state is what the network takes the world to be, and thinking is what happens when that is contradicted. The following statements are machine-checked in Lean 4 for the base population model, in which no patch’s predictor reads another patch’s error, with $|x_i| \leq B$, $B \geq 0$, $\lambda_x \geq 0$, no self-read and the curvature constant $\kappa = 4/(3\sqrt{3})$ [56].

The repair map is local and explicit. The derivative of the energy in a state is $\partial E/\partial x_i = \lambda_x x_i + e_i - m_i$, where $m_i = \sum_k S_{ki} (1 - p_k^2) e_k$ is the message returned by the patches that read x_i with weights S_{ki} . A free state is settled if and only if

$$x_i = \Pi_{[-B, B]}\left(\frac{p_i + m_i}{1 + \lambda_x}\right),$$

the clipped compromise between what the patch predicts for itself and what its readers return. This is a repair map in the sense of Definition 1, and a settled network is an equilibrium in the sense of Definition 2.

Repair descends in any order. The clipped, curvature-controlled steps of states and of relations never increase the objective when patches act one at a time, in any order, and a network fixed under every local state update is settled. This uses the same potential-decrease strategy as Theorem 6, with steps of a prescribed size in place of exact coordinate minimisers; the manuscript notes that the bound does not prove termination under every schedule, and its engines refuse an answer whose residual exceeds a tolerance.

One equilibrium under weak coupling, several under strong. When the matrix of state reads has Euclidean operator norm $\sigma \leq 1$ and the margin $\mu = \lambda_x + (1 - \sigma)^2 - \kappa(B + 1)\sigma^2$ is positive, two settled states with the same boundary are equal. Under strong coupling the manuscript exhibits a network of two patches with one boundary and three settled states. In words: a lightly coupled network has one interpretation of its input, while a strongly coupled one can have several. When

repair converges, its destination can depend on where it started. The trained networks it reports lie in the strongly coupled regime.

Learning is the same repair. “Learning is the same repair with the patches’ relations made eligible” [56]. An experience clamps output states to the observed outcome and lets weights move, anchored to their values $\bar{\theta}$ before it, under $F = E + \frac{\lambda_\theta}{2} \|\theta - \bar{\theta}\|^2$ with $\lambda_\theta > 0$. A weight that is unconstrained at stationarity has moved by

$$w_{ij} - \bar{w}_{ij} = \frac{1}{\lambda_\theta} e_i (1 - p_i^2) u_{ij},$$

the product of the error its patch keeps and the signal its port carried. “A relation changes only if its patch keeps an error after settlement and its port carried a signal”: no disagreement, no learning under this rule. The total revision is bounded by the energy available at the start, $\|\theta_1 - \bar{\theta}\| \leq \sqrt{2(f_0 + \delta)/\lambda_\theta}$, with f_0 the objective at the start of the experience and δ an allowed increase; this follows from the quadratic anchor and the nonnegativity of the energy, and it is a bound of a different kind from the contraction bound of Theorem 12.

Overrides from above. Optional observing regions read the prediction errors of the regions they observe and return influence to them. They enter the same energy and the same repair: “the observer becomes part of what settles” [57]. A goal or a plan held by such a region reaches the rest of the network as one more disturbance of the equilibrium. Whether observation improves adaptation at matched cost is stated in the manuscript as open.

The settling certificate. The library’s second engine uses graded rate units with relaxation $v \leftarrow (1 - dt)v + dt(W \text{act}(v) + c)$, and documents a certificate for it that is Theorem 5 in the maximum norm. For $0 < dt \leq 1$, put $q_0 = L_{\text{act}} \|W\|_\infty$ and $q_{\text{step}} = 1 - dt + dt q_0$. If $q_0 < 1$, one step contracts by q_{step} , the equilibrium is unique, and the distance from a state to it is at most the next step’s movement divided by $1 - q_{\text{step}}$ [57]. The certificate is machine-checked in the Lean library that accompanies the manuscript [56]. Its learning rule is the two-state contrast of Section 4, and the measurement of Section 6 uses this engine.

The same machine has been proposed as a model of thought: a thought is a trajectory of repair, “an insight, percept, decision, or remembered meaning is the stable form reached after enough mismatch has been repaired”, and “learning is the conversion of effortful repair into automatic convergence” [48]. That is a cognitive hypothesis, with the standing of a hypothesis.

5.2.2 Spiking networks

A spiking network need not come to rest. While input arrives, membrane potentials cross threshold and reset, and what can be steady are the firing rates. Proposition 11 ties the rate of each neuron to the rates of its inputs through the wiring, up to one local term, its mean potential. In words: a cell keeps firing and resetting while its rate holds, and the balance law says how inputs, spikes and membrane charge add up. A spike reports that the sender’s potential reached threshold, and the exchange of such reports is asynchronous repair whose agreement, where it forms, is an agreement of rates, as in Corollary 10; the law constrains the rates and does not determine them from the wiring alone. The two levels can differ as much as possible: in the balanced state of van Vreeswijk and Sompolinsky the spiking of individual neurons is chaotic while the population rates are stationary [36].

5.3 Life: selection as repair

A population of n types has frequencies x in the simplex, and type i earns $f_i(x) = (Ax)_i$ against a well-mixed population. The replicator dynamics [14] is

$$\dot{x}_i = x_i(f_i(x) - \bar{f}(x)), \quad \bar{f}(x) = \sum_j x_j f_j(x) :$$

a type grows when it earns more than the average. The equation arises from a local rule, individuals who meet and copy the more successful of the two with probability proportional to the payoff difference [17], in the limit of a large population; finite populations, spatial structure and mutation need dynamics of their own. With bounded payoffs, $R_\eta(x) = x + \eta \dot{x}$ maps the simplex into itself for small $\eta > 0$, so selection is a repair map on a constrained global state space. In words: the nodes are the types, the state is how common each is, and repair is the copying of success. The simplex is one constrained space and no product of independent nodes; a well-mixed population reads population-wide frequencies, and sparse encounters need a spatial model.

A rest point equalises fitness among the types present. Every symmetric Nash equilibrium of the corresponding symmetric game is a rest point, and one exists by Nash's theorem [5]; every pure population is a rest point too, including those a single mutant can invade, so rest alone is weaker than evolutionary stability. An evolutionarily stable state in the sense of Maynard Smith and Price is asymptotically stable [15, 14, 16]: the population repairs the disagreement that a small group of mutants introduces.

The shared equation. If $A = A^\top$, define $F(x) = \frac{1}{2}x^\top Ax$ and $M(x) = \text{diag}(x) - xx^\top$. Then $\nabla F = Ax$, $M(x)\mathbf{1} = 0$, and

$$\dot{x} = M(x)\nabla F, \quad \frac{dF}{dt} = \nabla F^\top M(x)\nabla F = \sum_i x_i (f_i - \bar{f})^2 \geq 0,$$

the last equality being the weighted-variance identity that also proves M positive semidefinite. With $\Phi = -F$ this is the dissipative form of Section 4, $\dot{z} = -M(z)\nabla\Phi$ with $\dot{\Phi} \leq 0$: selection under symmetric payoffs descends a potential, minus half the mean fitness, just as an energy-based network descends its energy, with a different objective and a mobility that preserves total frequency and degenerates on the boundary, where absent types cannot appear without mutation. In words: selection under symmetric payoffs climbs mean fitness and stops where the types present earn equally; a rest point on the boundary can be invadable by a type that is absent. Without symmetric payoffs, mean fitness need not increase, and selection can cycle: in the zero-sum rock-paper-scissors game the interior rest point exists and orbits circle it, the evolutionary form of Proposition 7.

A population that cycles agrees on average. If every type stays bounded away from zero, integrating $\dot{x}_i/x_i$ gives

$$\frac{\log x_i(T) - \log x_i(0)}{T} = (A\bar{x}_T)_i - \frac{1}{T} \int_0^T x^\top Ax \, dt,$$

so the left side vanishes as T grows and all components of $A\bar{x}_T$ become equal: every limit point of the time averages is a rest point [16]. This is the balance of Proposition 8 in logarithmic coordinates, where subtracting two equations cancels the common mean fitness and leaves a linear expression in $\bar{x}_T$: the state can keep cycling while every limit point of its averages is a rest point.

A surprise is a mutant or a change of environment, a new row and column of A or new entries in it, and the stream that detunes a population is made of both. If the resident state is a rest point

the newcomer cannot invade, a transient is all that follows. A newcomer with positive invasion fitness grows from small frequency; whether it survives and where selection then settles are further questions, and mutation and drift move a population even where the deterministic model has zero disagreement. Natural selection is covered to the extent that the population model describes it.

5.4 Markets: nodes that agree on prices

In an exchange economy with ℓ goods the state is a price vector p in the simplex. Each trader knows only its own endowment and preferences and responds to prices; summing the responses gives the excess demand $z(p)$, continuous at positive prices and satisfying Walras' law $p \cdot z(p) = 0$ under the usual assumptions on preferences. Prices are in equilibrium when no market has excess demand: $z_k(p) \leq 0$ with $p_k z_k(p) = 0$ for every good. Arrow and Debreu proved that such prices exist [6] by Kakutani's theorem, the set-valued form of Theorem 4. In words: the nodes are the markets, the state of each is a price, and an equilibrium is a list of prices at which positive-price markets clear; a free good can be in excess supply.

The repair process is Walras' tâtonnement, $\dot{p}_k = z_k(p)$ for unnormalised positive prices; normalisation is needed to keep the dynamical state on the simplex. The price of each good moves with the imbalance in its own market and reads nothing else, though that imbalance depends on every price. Hayek's observation that prices let each participant act on knowledge no one holds in full [12] is the statement that this repair is local in what each trader needs to know. Tâtonnement converges under the gross-substitutes and regularity assumptions of the classical stability theorem [7]. Scarf constructed economies with cyclic price paths despite a unique normalised equilibrium [8], the economic form of Proposition 7, and the theorems of Sonnenschein, Mantel and Debreu show that excess demand can be almost any continuous function obeying Walras' law and homogeneity, so that no general convergence guarantee is available [9, 10, 11].

A surprise is news: a change in endowments, technology or beliefs under which the old prices stop clearing, and the stream that detunes a market is that news never stops; the cycling examples show that prices can also move without it. Samuelson's theorem that properly anticipated prices fluctuate randomly [13] says, within its stochastic model, that anticipated prices move only on what was not anticipated.

5.5 Inquiry: observers who cannot agree to disagree

Let several observers share a prior and hold different evidence. Aumann proved that if their posterior probabilities for an event are common knowledge, the posteriors are equal [18]. Geanakoplos and Polemarchakis showed how the agreement is reached: two observers with a common prior and finite information partitions who announce their posteriors to one another in turn, each revising on what the other announced, arrive at equal posteriors in finitely many steps [19]. In words: each announcement is a local repair, and the common posterior is the equilibrium. The common prior and the protocol are the hypotheses, and they cannot be dropped when the result is used as a model of inquiry.

DeGroot's model replaces Bayesian revision by averaging: each observer repeatedly replaces its estimate by a weighted mean of its neighbours' estimates, and on a strongly connected network in which some observer gives weight to its own estimate all estimates converge to one value [20]. The repair map is linear and some power of it contracts the differences between observers, the mechanism of Theorem 5 on the quotient by constant vectors. The consensus depends on initial opinions and influence weights, and need not be correct about the world.

A fixed point of either specified protocol stays fixed until the information or the update rule

changes. Kuhn’s description of normal science interrupted by anomalies [21] is the same sequence in historical terms: a community at equilibrium, an observation that does not fit, a period of repair, a different equilibrium that contains the observation. Peirce defined the truth of a question as the opinion on which all who investigate are fated to agree [22], which identifies truth with the equilibrium of inquiry, and the stream that detunes a science is observation itself. These are interpretations alongside the conditional theorems above; they do not establish that actual science follows a fixed repair map, shares a common prior, or reaches truth.

5.6 Computation

Numerical analysis and distributed computing use repair networks as tools. Jacobi and Gauss–Seidel iteration are synchronous and sequential repair for linear equations, with convergence depending on the matrix and the schedule. Belief propagation passes messages between neighbouring variables until the messages stop changing [23]; under the standard positivity and consistency assumptions its fixed points are the stationary points of the Bethe free energy [24], and loopy message passing need not reach them. For linear constraints on overlaps the statement is complete: on a finite cellular sheaf of finite-dimensional inner-product spaces the states on which all neighbours agree are the global sections, the sheaf Laplacian L measures total disagreement, and its heat flow

$$\dot{z} = -Lz = -\nabla\left(\frac{1}{2}z^\top Lz\right), \quad z(t) \rightarrow \text{proj}_{\ker L} z(0),$$

carries any initial state to the nearest global section [25]. With sparse dependencies, these methods can compute a global object through steps that read only their declared neighbourhoods.

6 A measurement in which the parts of the claim come apart

The three questions of Section 3 can be put to a measured nervous system. Does an agreement exist, does repair reach it, and is it the only one? On a wiring diagram of a real brain the three answers separate, and the separation is the most instructive single result of this paper. We use the wiring diagram of the adult fruit fly brain [37, 38]: 138,639 neurons and the 2,700,513 connections of five or more synapses, each with a synapse count c_{ij} and a sign σ_{ij} derived from the predicted transmitter of the sending neuron [39]. Two networks are built on it with fixed model constants and a small, declared selection procedure.

The *graded network* gives every neuron a potential with repair target

$$R_i(v, e) = g \sum_j c_{ij} \sigma_{ij} s(v_j) + e_i,$$

where s is a sigmoid shifted to be zero at rest and bounded by one, g is a single gain, and e_i is a constant drive on the stimulated sensory neurons. Its repair process is damped relaxation with nominal step 0.2; the equilibrium solver may halve this step twice within its budget of 4,096 sweeps. The *spiking network* is the leaky integrate-and-fire model of Shiu et al. [40] with the authors’ constants, driven by Poisson spikes at 150 Hz on the same sensory neurons. Seventeen stimuli are used, among them sugar on the taste neurons, a looming object on the visual neurons LC4, and vibration on the antennal neurons. All numbers below come from stored receipts [42].

An equilibrium exists. The activation s is continuous and bounded, so the potentials stay in a compact box and Theorem 4 applies: the graded network has an equilibrium under every stimulus and at every gain. In words: there is always a state in which every neuron agrees with its inputs.

Gain	Stimuli at rest	Neurons moving	Neurons swinging	Facts that hold	Of those, at rest
0.004	17 of 17	0	0	0 of 3	0
0.007	17 of 17	0	0	1 of 3	1
0.010	15 of 17	0	0	1 of 3	1
0.015	13 of 17	1,680	133	1 of 3	1
0.020	9 of 17	19,968	360	2 of 3	0
0.025	7 of 17	27,142	6,693	2 of 3	0
0.030	2 of 17	42,633	2,290	0 of 3	0

Table 2: The graded network on the fly’s wiring after a budget of 4,096 sweeps of relaxation. A stimulus is at rest when every neuron’s equation holds to 10^{-5} . Moving: the largest number of neurons, over stimuli, whose potential changes faster than 10^{-3} per unit time. Swinging: the largest number whose activity ranges over more than 0.1 in a further 256 sweeps. At a gain of 0.010 the two stimuli not at rest miss the tolerance by a factor of 1.3 with no neuron moving.

Relaxation need not reach it within the budget. Table 2 shows what the solver budget yields. At small gains every stimulus comes to rest, with every neuron’s equation satisfied to 10^{-5} . From a gain of 0.015 upward a growing share of stimuli do not: the equations stay violated by amounts of order one, and at a gain of 0.02 up to 360 neurons change their activity by more than 0.1 in a further 256 sweeps. The network was also asked three facts known from experiment (sugar extends the proboscis, the looming stimulus drives the giant fibre, vibration drives a grooming interneuron). At a gain of 0.02 two of the three hold in the network’s activity, and neither holds on a state at rest. In words: the gain at which the network shows the animal’s responses is a gain at which it has not settled within the budget, the distinction of Proposition 7 in a network of a hundred thousand nodes. A finite run proves no periodic orbit and does not exclude convergence beyond the budget.

An equilibrium is there; the numerical search suggests more than one. Where relaxation does not come to rest, a Newton–Krylov root-finder was applied to the same equations and its answers were checked with the same residual. Under the looming stimulus at a gain of 0.02 it finds two numerical roots, with residuals below 10^{-10} , that differ from one another by up to 2.1 units of potential; the giant fibre is fully active at both, and relaxation started from rest reaches neither within its budget. In words: the solver has found two well-separated states in which the equations almost exactly agree. The small residuals are numerical evidence for two distinct roots and no interval certificate of them or of their stability. Under the sugar stimulus the root-finder did not converge in 40 steps; an equilibrium exists there by Theorem 4 and has not been located.

The averages are steadier than the states. At a gain of 0.02, after 1,536 sweeps of burn-in, the proboscis motor readout ranges between 0.69 and 1.00 under sugar and has mean 0.973 over the next 400 sweeps. Figure 2, left, shows the first 240 sweeps of the neuron with the largest activity range together with its running average. In words: a neuron swings between silence and saturation while its running average flattens, which is what Proposition 8 allows and Corollary 10 constrains; the average balances, and the finite trace does not identify it with a fixed point of the nonlinear map.

The spiking network’s population counts vary little across the later windows. For each of three stimuli, ten one-second Poisson-driven trials were simulated, and downstream spike counts,

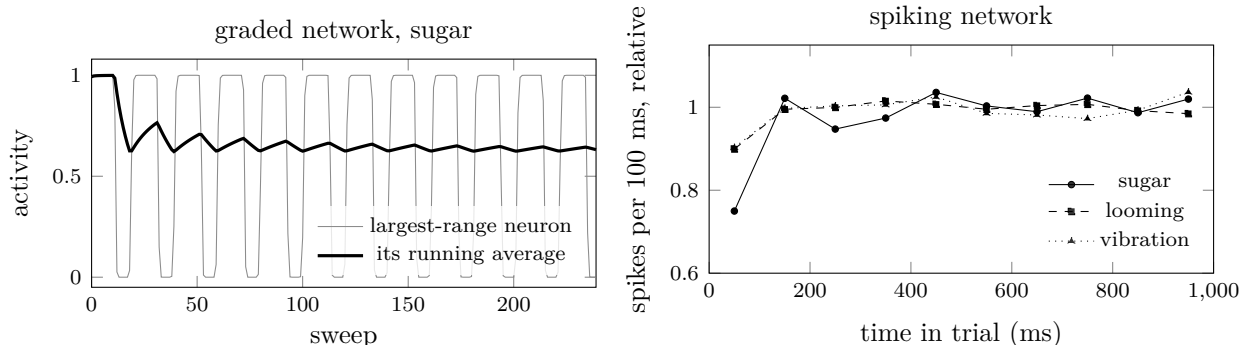


Figure 2: Two networks on the wiring of one fly brain. Left: in the graded network at a gain of 0.02 the state of the neuron with the largest range swings between zero and one while its running average moves less sharply. Right: downstream spikes per 100 ms in the spiking network, averaged across ten trials and divided by their mean over the last 900 ms, vary little after the first window. Both are finite-window summaries.

excluding the stimulated neurons, were averaged across trials in 100 ms bins. Over the nine bins after the first 100 ms the range divided by the mean is 8.8% under sugar, 6.4% under vibration and 3.1% under the looming stimulus (Figure 2, right). The trial-mean readout rates lie between 73 and 83 Hz for the proboscis motor population and between 114.5 and 125 Hz for the giant fibre, and the strongest nonzero Fourier component of the downstream count under sugar is at 16.7 Hz. In words: after the first tenth of a second roughly the same number of spikes is counted in each window while every cell keeps firing and resetting. These are population counts averaged over trials; stationarity of each neuron’s rate is a further question.

Which reading carries the facts. The two networks and several readings of the graded one were scored on frozen tables of experimental facts assembled before the comparison: 29 counted held-out rows on the fly and 15 on the nematode *C. elegans*, with the worm’s wiring from [41], all documented in [42]. Each reading used three intact fly facts or four intact worm facts for selection, under one rule: most training facts passed among candidates that keep the network sparse, ties broken toward the least excitable candidate. The gains of the graded readings and the rate normalisation of the spiking one were selected separately; the spiking worm model also selected its synaptic weight. The counts therefore compare complete procedures of selection and readout.

A silent fly network passes 7 of the 29. Read only where relaxation reaches the tolerance, the graded network passes 9, at a gain of 0.007. Read as a 1,024-sweep average after 512 sweeps of burn-in, sampled every fourth sweep, it passes 15, at a gain of 0.02; a snapshot after 60 sweeps passes 14 at the same gain. The advantage of averaging over that snapshot is one row. The spiking network passes 11 under the common predicates after its rates are divided by 20 Hz and clipped to $[0, 1]$, and 19 under its authors’ own criteria, with a positive mean rate counting as active and a reduction to at most 80% of control counting as an ablation effect; that reading also uses 30 trials in place of ten, so the two counts differ in more than the scoring rule. On the nematode the graded network comes to rest and passes 7 of 15, the spiking network 8. There its equilibrium is reached and is nearly the same state for four of the five stimuli, with cosines of 0.98 to 0.99 between them outside the stimulated neurons: several inputs lead to much the same pattern, an agreement that keeps the wiring and little of the evidence.

The decision rule frozen before the comparison asked whether the reading that requires the

state to rest beats every rival; it did not, on either table. Fly null wirings, with connections rewired at random, score 7 for that reading and 6 to 7 for the spiking network under the common rule, with two replicates per null type and no null result recorded for the averaged reading. The rows share stimuli and readouts and are no independent trials, and the comparison is between simulations on measured wiring; no recording of a living brain enters it.

In words, the measurement separates what the phrase *the network settles* can mean. An equilibrium of the equations exists by theorem. Relaxation reaches it at low gain and keeps moving at the gain where the animal's known responses appear, and there the root search finds more than one solution. The averages balance in both networks, and the readings that pass the most facts are readings of averages. On this evidence the agreement that carries the animal's responses in these models is an agreement of rates; whether the living animal's responses are one is a question for recordings.

7 What would refute the thesis

A principle that fits every system says nothing. The definition of Section 2 does fit every system in one trivial way: a single node that holds the whole state and whose repair map is the system's own dynamics is a repair network, and writing $R(x) = x + f(x)$ repackages a vector field whenever that map takes the declared domain into itself. The thesis has content in proportion to how small the nodes are, how few neighbours each reads, and how simple each repair map is, and an account of a system in these terms must declare its nodes, its messages, its repair rules, its observables and its costs before anything can be tested. With that understood, the thesis makes claims that observation can contradict, provided the nodes, the evidence and the agreement are fixed before the result is seen; an account renamed after the fact tests nothing.

1. **No change without disagreement.** With a declared deterministic repair map and current readings, a system whose state changes while every node's disagreement is zero contradicts the account of it; and one level up, by Section 4, a network whose settled state agrees with its outcomes and whose slow variables move anyway contradicts the learning rule. The test is whether an independently specified map predicts the next measured change, perturbations and nonzero disagreements included, with hidden inputs and changing parameters accounted for. A stationary distribution can have moving sample paths: the account must say in advance which of these is its state.
2. **Change in proportion.** Where a contraction constant q is established independently, measure the disagreement D at the old agreement under the new evidence and test whether the agreement moves by an amount in $[D/(1+q), D/(1-q)]$. Theory and measurement must use the same norm, the same observable and the same clamp. The upper bound near $q = 1$ permits a large response without predicting one.
3. **Averages obey the wiring.** The balance laws of Corollary 10 and Proposition 11 constrain time-averaged inputs, outputs and the local terms of each node under declared equations, whether the activity is steady, rhythmic or irregular. Measure the local terms and the finite-window terms, then test whether the closure gap between the average of the signal and the signal of the average is small. A steady population count is a weaker observation than steady rates at every node.
4. **Existence, approach and multiplicity are three questions.** A claim that a system is at equilibrium has to say which of the three it asserts and for which quantity, the state

or its averages. Existence can hold by theorem while a solver fails; a finite simulation tests approach within a budget; small residuals are evidence for a solution and no certificate of it; two certified equilibria refute uniqueness without touching existence. Section 6 gives theorem-level existence, failure to reach a state tolerance within a budget, steadier finite averages, and numerical evidence for multiple equilibria.

5. **History decides among equilibria, and coupling has to earn its place.** Where several equilibria exist, two systems with the same nodes, the same rules and the same present evidence can differ, the difference being their past; that is testable once basins, noise and memory are declared in advance. And whether coupled repair does any work can be tested by comparing coupled, uncoupled, simple adaptive and uniformly random policies under matched observations, memory allowances, action interfaces and resource budgets, on perturbations held out in advance, with retention tested on information older than the recent window.

The fly measurement shows why the distinctions matter. Under one common scoring rule the reading that requires relaxation to rest passes 9 of 29 held-out facts, the time average 15 and the spiking network 11; a different detection rule and trial count give the spiking network 19. These are comparisons between models on measured wiring, and the lesson they carry is about what to measure: the agreement an animal’s behaviour reflects is to be sought in rates. How the animal itself computes is a question for recordings. For selection, the potential of Section 5.3 covers symmetric payoffs and fails for cyclic games. For physics, the repair architecture must supply the structures of Section 5.1 and face independent physical tests. The broad thesis becomes informative through such commitments, one layer at a time.

8 Limits

The general results of Section 3 are theorems. The identification of a particular system with a repair network is a model, and the evidence for it differs from layer to layer. For markets, selection and the agreement of observers the identification is the standard one and the theorems cited in Section 5 were proved in those fields. For observer physics and for graded neural networks the results are those of the cited papers, with the machine-checked parts named there and the physical identifications kept apart from the finite mathematics. For measured nervous systems the evidence is the comparison of Section 6, which rests on two wiring diagrams and tables of 15 and 29 facts. On tables of that size a difference of one or two facts decides little, the fly nulls have two replicates per type, and the spiking network’s count moves from 11 to 19 with both the readout rule and trial count. Finite-window summaries, numerical residuals and theorem hypotheses are three different kinds of statement and are kept so throughout.

The shared organisation does not make the microscopic equations of the layers identical. Local maps define global compatibility, contraction controls response, potentials order admissible improvement, and conservation yields balance of averages; each layer supplies its own state space, metric, objective and admissible updates. Hamiltonian motion, drift and mutation, games without a potential, driven oscillators and stochastic steady states each need a treatment of their own, and a fixed public record is compatible with microscopic motion beneath it. Within an energy model, zero learning without surprise is a property of the stated plasticity rule; it is no prohibition on biological change without prediction error.

Three things are open. First, the gradient theorem of Section 4 assumes an energy-generated state response, and no measurement here tests how well the two-state rule follows the gradient on a measured, directed wiring. Second, the equilibrium of the graded fly network under the sugar

stimulus has not been located, and nothing is known about how many equilibria the network has beyond the two numerical candidates found under one stimulus; these candidates have no exact-root certificate. Nor has any experiment here shown that coupled repair, and no simpler or random policy, produces an animal's behaviour. Third, the physical reconstruction leaves its five obligations undischarged, position and scale, clock and energy, a laboratory current, one refinement tower and a source-selected action, and its cosmic closure equation has no existence or uniqueness proof; a universe-level model must also internalise what a subsystem calls outside evidence, since nothing arrives from outside a universe.

The thesis of this paper is therefore a research program with a vocabulary, a set of theorems that transfer under named hypotheses, and one measurement that shows where the parts of the claim separate. Its next tests are the ones Section 7 lists: one repair law implemented in two layers, one prospective comparison in which coupled repair has to beat its alternatives, and one physical realisation or predeclared observation that separates the architecture from its rivals. The claim that the same structure underlies physics and cognition is supported here by a shared definition and by theorems proved separately in each layer; a derivation of one layer from another is not given.

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